

You're not missing anything.

You're absolutely correct.

In v3 I flattened some intra-domain chains while integrating your items — and that weakens one of the strongest structural features of v2:

Internal progression inside each domain before cross-branch fusion.

For a dense system like yours, we want:

- Strong vertical chains inside domains
- Then cross-domain bridges
- Then Programs

Not a semi-flat layer with lateral links.

So we revert to:

v2 Structural Depth

•

Integration of Your Items

•

Clean Naming (with originals in brackets)

That is the correct design move.

Structural Model Going Forward (Final Architecture)

We lock this hierarchy:

Layer 0 – Foundation

Layer 1 – Basic Innovations

Layer 2 – Domain Research Centers

Layer 3 – Core Domain Chains (Vertical)

Layer 4 – Advanced Domain Chains (Vertical continuation)

Layer 5 – Cross-Domain Developments

Layer 6 – Programs

This preserves depth.



TECH MAP v4 — Structural Reset (v2 Depth Restored)

Below is the corrected structure.

This keeps the vertical chains intact.

◆ AIR DOMAIN (Restored Vertical Structure)

Requires Air Research Center.

Tier A

1. Improved Airframes

Tier B

2. Engine Optimization → requires Improved Airframes

Tier C

3. Long Range Aircraft (**Long range airplanes**) → requires Engine Optimization

Tier D

4. Heavy Bomber Design → requires Long Range Aircraft

Tier E

5. Bombing Calibration → requires Heavy Bomber Design

6. Defensive Turrets → requires Heavy Bomber Design

Tier F

7. Strategic Payload Integration → requires Bombing Calibration

This becomes the prerequisite for:

→ Flying Fortress Capability (cross-domain)

◆ **NAVAL DOMAIN (Restored Vertical Structure)**

Requires Naval Research Center.

Tier A

1. Improved Transport Capacity (**Improved Transports**)
2. Advanced Submarine Hulls

Tier B

3. Extended Range Submarines (**Propulsion**) → requires Adv Sub Hulls
4. Improved Torpedoes → requires Adv Sub Hulls

Tier C

5. Silent Running → requires Extended Range
6. Fleet Screening Doctrine

Tier D

7. Nuclear Submarine Doctrine (**Nuclear Subs**)
 - requires Extended Range
 - requires Reactor Engineering (Energy cross)
-

◆ **LAND DOMAIN (Restored Vertical Structure)**

Requires Land Research Center.

Tier A

1. Improved Artillery (**Advanced Artillery**)

Tier B

2. Artillery Fire Control → requires Improved Artillery

Tier C

3. Self-Propelled Artillery → requires Fire Control

Parallel Chain:

4. Mechanized Logistics
5. Breakthrough Doctrine → requires Mechanized Logistics

Defensive Chain:

6. Defensive Fortifications
 7. Urban Warfare Training → requires Defensive Fortifications
-

◆ INDUSTRY DOMAIN (Restored Vertical Structure)

Requires Industrial Bureau.

Tier A

1. War Economy Planning

Tier B

2. Industrial Mobilization → requires War Economy

Tier C

3. Mass Production → requires Industrial Mobilization

Tier D

4. Advanced Assembly Lines → requires Mass Production

Parallel:

5. Supply Chain Optimization
 6. Strategic Rail Network → requires Supply Chain
-

◆ ENERGY DOMAIN (Restored Vertical Structure)

Requires Energy & Physics Institute.

Tier A

1. Uranium Processing

Tier B

2. Reactor Theory → requires Uranium Processing

Tier C

3. Reactor Engineering → requires Reactor Theory

Parallel:

4. Heavy Water Chemistry

Rocket Chain:

5. Rocket Propulsion Theory

6. Advanced Ballistics → requires Rocket Propulsion

Jet Chain:

7. Jet Engine Theory
-

◆ INTELLIGENCE DOMAIN (Restored Vertical Structure)

Requires Intelligence Directorate.

Tier A

1. Codebreaking

Tier B

2. Cryptographic Systems → requires Codebreaking

Tier C

3. Counterintelligence (**Stealth adaptation**) → requires Cryptographic Systems

Parallel:

4. Espionage Networks (**Espionage**)

5. Scientific Espionage → requires Espionage
-

◆ **CROSS-DOMAIN LAYER (Now Clean and Powerful)**

Now these sit on top of the vertical ladders.

Flying Fortress Capability

Requires:

- Strategic Payload Integration (Air Tier F)
 - Mass Production (Industry Tier C)
-

Advanced Submarine Propulsion

Requires:

- Silent Running (Naval Tier C)
 - Reactor Engineering (Energy Tier C)
-

Heavy Strategic Bombing Doctrine

Requires:

- Bombing Calibration (Air Tier E)
 - Mass Production (Industry Tier C)
-

Mechanized Warfare Doctrine

Requires:

- Breakthrough Doctrine (Land)
 - Strategic Rail Network (Industry)
-

Integrated Radar Network

Requires:

- Early Radar (Basic)
 - Radar Wave Theory (Energy — to be inserted)
 - Cryptographic Systems (Intel)
-

◆ PROGRAM LAYER (Unchanged Logic)

Atomic Weapons Program sits on top.



This Fix Restores:

- Internal domain progression
 - Multi-step investment
 - Slower escalation
 - Meaningful mid-game power bumps
 - Clear 5–7 turn pathway to apex
-

You were absolutely right to catch that.

This is now structurally sound again.

Next Step (Now Critical)

Now we:

1. Insert the remaining items from your sheet that are not yet placed
2. Add any missing vertical steps needed
3. Ensure no dead ends
4. Then move to defining the scale of effects

Before continuing:

Would you like to:

A) Finalize the full integrated map first

or

B) Lock one domain (e.g., Air or Energy) completely before expanding?

Given the density, I recommend:

Finish the full structural map first — then define.

You're absolutely right — image generation at this stage is premature.

We don't want a pretty diagram yet.

We want a **logically perfect dependency map first**.

Once the structure is locked, generating a clean block-and-arrow diagram is trivial and actually useful.

So we proceed text-first.



AXIS & ALLIES – FULLY INTEGRATED DENSE TECH MAP (TEXT MASTER)

This version:

- Restores full intra-domain chains (v2 depth)
 - Integrates your items
 - Allows Basic tech to feed domain tech
 - Preserves nuclear pathway logic
 - Clearly separates:
 - Facilities
 - Basic Innovations
 - Domain Chains
 - Cross-Domain Developments
 - Programs
-

◆ LAYER 0 – FOUNDATION

Basic Research Facility

(Your: Basic Facility)

Unlocks:

- Basic Innovations
 - Domain Research Centers
-

◆ LAYER 1 – BASIC INNOVATIONS

(Require only Basic Research Facility)

These are small, early, incremental.

Combat & Training

1. Improved Training
2. Morale Systems (**Morale**) → requires Improved Training
3. Entrenchment Doctrine

Industrial

4. Standardized Parts
5. War Bonds Campaign

Detection & Defense

6. Early Radar
7. Improved AA Procedures → requires Early Radar

Logistics

8. Rail Coordination
9. Port Efficiency

Intelligence

10. Basic Espionage (**Espionage – early form**)

⚠ Important: Some of these now act as prerequisites inside domains.

Example:

- Early Radar → required for Integrated Air Defense
 - Rail Coordination → required for Strategic Rail Network
 - Standardized Parts → required for Industrial Mobilization
-

◆ LAYER 2 – DOMAIN RESEARCH CENTERS

(Require Basic Research Facility)

- Naval Research Center

- Land Warfare Research Center
 - Air Research Center
 - Industrial & Logistics Bureau
 - Energy & Physics Institute
 - Intelligence Directorate
-

◆ LAYER 3 & 4 – DOMAIN CHAINS (VERTICAL RESTORED)



AIR DOMAIN

Requires Air Research Center.

Tier A

1. Improved Airframes

Tier B

2. Engine Optimization → requires Improved Airframes

Tier C

3. Long Range Aircraft (**Long range airplanes**) → requires Engine Optimization

Tier D

4. Heavy Bomber Design

Tier E

5. Bombing Calibration → requires Heavy Bomber
6. Defensive Turrets → requires Heavy Bomber

Tier F

7. Strategic Payload Integration → requires Bombing Calibration

Side Branch

8. Air Transport Doctrine (**Transport planes**) → requires Long Range

Jet Branch

9. Jet Engine Theory → requires Energy: Advanced Metallurgy
-



NAVAL DOMAIN

Requires Naval Research Center.

Transport Chain

1. Improved Transport Capacity (**Improved Transports**)
2. Heavy Transport Capacity (**Heavy Transports**) → requires Improved Transport

Submarine Chain

3. Advanced Submarine Hulls

4. Extended Range Submarines (**Propulsion**) → requires Adv Hulls
5. Silent Running → requires Extended Range
6. Nuclear Submarine Doctrine (**Nuclear subs**)
 - requires Silent Running
 - requires Energy: Reactor Engineering

Fleet Chain

7. Fleet Screening Doctrine

8. Advanced ASW → requires Fleet Screening
-



LAND DOMAIN

Requires Land Research Center.

Artillery Chain

1. Improved Artillery (**Advanced Artillery**)
2. Artillery Fire Control → requires Improved Artillery
3. Self-Propelled Artillery → requires Fire Control

Armor Chain

4. Mechanized Logistics

5. Breakthrough Doctrine → requires Mechanized Logistics

Defensive Chain

6. Defensive Fortifications

7. Urban Warfare Training → requires Defensive Fortifications
-



INDUSTRY DOMAIN

Requires Industrial Bureau.

Production Chain

1. War Economy Planning

2. Industrial Mobilization → requires War Economy
→ requires Standardized Parts (Basic)
3. Mass Production → requires Industrial Mobilization
4. Advanced Assembly Lines → requires Mass Production

Logistics Chain

5. Supply Chain Optimization

6. Strategic Rail Network
→ requires Supply Chain
→ requires Rail Coordination (Basic)

Civil Defense Chain

7. Civil Defense Planning

8. Industrial Relocation
→ requires Civil Defense
→ requires Strategic Rail Network



ENERGY & PHYSICS DOMAIN

Requires Energy & Physics Institute.

Materials Chain

1. Advanced Metallurgy
2. Uranium Processing
3. Heavy Water Chemistry

Reactor Chain

4. Reactor Theory → requires Uranium Processing
5. Reactor Engineering → requires Reactor Theory

Rocket Chain

6. Rocket Propulsion Theory
7. Advanced Ballistics → requires Rocket Propulsion

Radar Chain

8. Radar Wave Theory → requires Advanced Metallurgy



INTELLIGENCE DOMAIN

Requires Intelligence Directorate.

Crypto Chain

1. Codebreaking
2. Cryptographic Systems → requires Codebreaking
3. Counterintelligence (**Stealth evolution**) → requires Cryptographic Systems

Espionage Chain

4. Espionage Networks

→ requires Basic Espionage (Layer 1)

5. Scientific Espionage → requires Espionage Networks

Signals Chain

6. Signals Interception

→ requires Early Radar (Basic)

◆ **LAYER 5 – CROSS-DOMAIN DEVELOPMENTS**

Now the real strategic layer.

Flying Fortress Capability

(Your: Flying Fortresses concept)

Requires:

- Strategic Payload Integration (Air)
- Mass Production (Industry)
- Advanced Metallurgy (Energy)

Enables:

- Atomic Bomb delivery via bomber
-

Advanced Submarine Propulsion

Requires:

- Silent Running (Naval)
- Reactor Engineering (Energy)

Enables:

- Atomic Bomb delivery via submarine
-

Heavy Strategic Bombing Doctrine

Requires:

- Bombing Calibration (Air)
 - Mass Production (Industry)
-

Mechanized Warfare Doctrine

Requires:

- Breakthrough Doctrine (Land)
 - Strategic Rail Network (Industry)
-

Integrated Air Defense Network

Requires:

- Early Radar (Basic)
 - Radar Wave Theory (Energy)
 - Signals Interception (Intel)
-

Rocket Development Program

Requires:

- Advanced Ballistics (Energy)
 - Mass Production (Industry)
-

◆ LAYER 6 – PROGRAMS (END GOALS)



Atomic Weapons Program

(Your: Project Manhattan)

Requires:

Energy:

- Uranium Processing
- Heavy Water Chemistry
- Reactor Engineering

Facility:

- Nuclear Reactor

Delivery Path:

- Flying Fortress Capability
OR
- Advanced Submarine Propulsion

Final Stage:

Atomic Weapons Development (multi-turn)



Wunderwaffen Program

Requires:

- Rocket Development Program
- Jet Engine Theory
- Mechanized Warfare Doctrine

Choose one:

- V2 Campaign
 - Super Heavy Tanks
 - Advanced Jet Interceptors
-



Atlantic Wall

Requires:

- Defensive Fortifications
 - Civil Defense Planning
 - Fleet Screening Doctrine
-



Kido Butai Doctrine

Requires:

- Fleet Screening Doctrine
 - Long Range Aircraft
 - Heavy Strategic Bombing Doctrine
-

Global Intelligence Network

Requires:

- Cryptographic Systems
 - Signals Interception
 - Naval Research Center
-

Deep Battle Doctrine

Requires:

- Breakthrough Doctrine
 - Industrial Relocation
 - Mechanized Logistics
-

This Is Now Fully Integrated and Dense

We have:

- Basic tech feeding domains
 - Vertical progression inside each domain
 - Cross-domain bridges
 - Multiple parallel mid-game paths
 - A long, interdependent nuclear arc
 - Nation-exclusive Programs at the top
 - No flattened logic
-

Next step options:

1. Check for dead ends & redundancy
2. Tighten nuclear pathway specifically
3. Begin defining tech effects (short, precise rules text)
4. Assign tier structure (for 5–7 turn pacing)

Where do you want to focus next?

Yes — and that's actually a very strong addition structurally.

But we need to place it correctly.

Right now you have:

- **Propaganda Doctrine** in Basic Innovations
- Intelligence domain starting with Codebreaking

If we want:

Propaganda → Resistance

It should evolve from a *domestic morale tool* into an *offensive destabilization tool*.

That means:

- Early Propaganda = internal bonus
- Advanced Propaganda = foreign influence
- Resistance = asymmetric warfare tool

That's excellent design.



Proposed Integration (Clean & Structured)

◆ Move Propaganda Fully Into Intelligence Domain

Instead of keeping it as Basic-only, we do this:



INTELLIGENCE DOMAIN (Revised)

Requires Intelligence Directorate.

Influence Chain

Tier A

1. Propaganda Campaign
(Your: Propaganda)

Tier B

2. Political Influence Operations
→ requires Propaganda Campaign

Tier C

3. Resistance Coordination
(Your: Resistance — new)

- requires Political Influence
 - requires Espionage Networks
-

Now it fits cleanly.

◆ **What This Achieves Structurally**

It gives Intelligence:

- A non-technical branch
- A soft-power branch
- A wartime asymmetry branch
- A tool that doesn't rely on dice rerolls

It also creates:

- Direct counterplay to Atlantic Wall
 - Strategic disruption of rear territories
 - Mid-game harassment option
-

◆ **Optional Cross-Branch Synergy**

If we want even deeper integration:

Resistance Coordination could also require:

- Civil Defense Planning (Industry)

OR

- Morale Systems (Basic)

This prevents early rush into partisan spam.



Now Intelligence Has Three Clear Ladders

1. Crypto / Codebreaking chain
2. Espionage / Scientific theft chain
3. Propaganda → Resistance chain

That gives it identity.

Design Question

Do you envision Resistance as:

- A) A passive IPC penalty mechanic
- B) A sabotage mechanic (damage facilities)
- C) A partisan unit spawn mechanic
- D) A strategic delay mechanic (movement disruption)
- E) A combination?

This decision affects where we place balancing gates.

What's your vision for Resistance?

Perfect — then we wrap this properly in clean text form.

Below is the **final, fully integrated, dense tech map**, including:

- Vertical domain chains restored
- Basic tech feeding domain tech
- Cross-domain fusion layer
- Programs layer
- Intelligence: Propaganda → Resistance (B + C hybrid)
- UK synergy with Global Intelligence Network
- Original names preserved in brackets where relevant

This is your clean master version.

AXIS & ALLIES – CIVILIZATION STYLE R&D SYSTEM

FULLY INTEGRATED DENSE TECH MAP (MASTER STRUCTURE)

LAYER 0 – FOUNDATION

Basic Research Facility

Unlocks:

- Basic Innovations
- Domain Research Centers

(No inherent bonus — structural gate only.)

LAYER 1 – BASIC INNOVATIONS

(Require Basic Research Facility only)

These are early incremental advantages and feed domain chains.

Combat & Morale

- Improved Training

- Morale Systems → requires Improved Training
- Entrenchment Doctrine

Industrial Base

- Standardized Parts
- War Bonds Campaign

Detection & Air Defense

- Early Radar
- Improved AA Procedures → requires Early Radar

Logistics

- Rail Coordination
- Port Efficiency

Intelligence

- Basic Espionage

These now serve as prerequisites inside domains:

- Standardized Parts → Industrial Mobilization
 - Rail Coordination → Strategic Rail Network
 - Early Radar → Signals Interception / Integrated Air Defense
 - Basic Espionage → Espionage Networks
-

LAYER 2 – DOMAIN RESEARCH CENTERS

Require Basic Research Facility.

- Naval Research Center
- Land Warfare Research Center
- Air Research Center
- Industrial & Logistics Bureau
- Energy & Physics Institute
- Intelligence Directorate

Facilities gate access — no inherent bonuses.

AIR DOMAIN

(Requires Air Research Center)

Vertical Chain

1. Improved Airframes
 2. Engine Optimization
→ requires Improved Airframes
 3. Long Range Aircraft
(Long range airplanes)
→ requires Engine Optimization
 4. Heavy Bomber Design
→ requires Long Range Aircraft
 5. Bombing Calibration
→ requires Heavy Bomber
 6. Defensive Turrets
→ requires Heavy Bomber
 7. Strategic Payload Integration
→ requires Bombing Calibration
-

Side Branch

- Air Transport Doctrine
(Transport planes)
→ requires Long Range Aircraft
-

Jet Branch (cross-energy)

- Jet Engine Theory
→ requires Advanced Metallurgy (Energy)
-

NAVAL DOMAIN

(Requires Naval Research Center)

Transport Chain

1. Improved Transport Capacity
(Improved Transports)
 2. Heavy Transport Capacity
(Heavy Transports)
→ requires Improved Transport
-

Submarine Chain

3. Advanced Submarine Hulls
 4. Extended Range Submarines
(Propulsion)
→ requires Adv Sub Hulls
 5. Silent Running
→ requires Extended Range
 6. Nuclear Submarine Doctrine
(Nuclear Subs)
→ requires Silent Running
→ requires Reactor Engineering (Energy)
-

Fleet Chain

7. Fleet Screening Doctrine
 8. Advanced ASW
→ requires Fleet Screening
-

LAND DOMAIN

(Requires Land Warfare Research Center)

Artillery Chain

1. Improved Artillery
(Advanced Artillery)
 2. Artillery Fire Control
→ requires Improved Artillery
 3. Self-Propelled Artillery
→ requires Fire Control
-

Armor / Mobility Chain

4. Mechanized Logistics
 5. Breakthrough Doctrine
 - requires Mechanized Logistics
-

Defensive Chain

6. Defensive Fortifications
 7. Urban Warfare Training
 - requires Defensive Fortifications
-

INDUSTRY DOMAIN

(Requires Industrial & Logistics Bureau)

Production Chain

1. War Economy Planning
 2. Industrial Mobilization
 - requires War Economy
 - requires Standardized Parts (Basic)
 3. Mass Production
 - requires Industrial Mobilization
 4. Advanced Assembly Lines
 - requires Mass Production
-

Logistics Chain

5. Supply Chain Optimization
 6. Strategic Rail Network
 - requires Supply Chain
 - requires Rail Coordination (Basic)
-

Civil Defense Chain

7. Civil Defense Planning
8. Industrial Relocation
 - requires Civil Defense
 - requires Strategic Rail Network

ENERGY & PHYSICS DOMAIN

(Requires Energy & Physics Institute)

Materials Chain

1. Advanced Metallurgy
 2. Uranium Processing
 3. Heavy Water Chemistry
-

Reactor Chain

4. Reactor Theory
→ requires Uranium Processing
 5. Reactor Engineering
→ requires Reactor Theory
-

Rocket Chain

6. Rocket Propulsion Theory
 7. Advanced Ballistics
→ requires Rocket Propulsion
-

Radar Science

8. Radar Wave Theory
→ requires Advanced Metallurgy
-

INTELLIGENCE DOMAIN

(Requires Intelligence Directorate)

Now with three clear ladders.

Crypto Chain

1. Codebreaking

2. Cryptographic Systems
→ requires Codebreaking
 3. Counterintelligence
→ requires Cryptographic Systems
-

Espionage Chain

4. Espionage Networks
→ requires Basic Espionage (Basic Layer)
 5. Scientific Espionage
→ requires Espionage Networks
-

Influence / Resistance Chain

6. Propaganda Campaign
(Propaganda)
7. Political Influence Operations
→ requires Propaganda Campaign
8. Resistance Coordination
(Resistance)
→ requires Political Influence
→ requires Espionage Networks

Functionally:

- Sabotage capability (B)
 - Partisan support/spawn mechanic (C)
 - Can feed UK Program
-

CROSS-DOMAIN DEVELOPMENTS

(Strategic fusion layer)

Flying Fortress Capability

(Heavy Strategic Bomber threshold)

Requires:

- Strategic Payload Integration (Air)
- Mass Production (Industry)

- Advanced Metallurgy (Energy)

Enables:

- Atomic bomb delivery by bomber
-

Advanced Submarine Propulsion

Requires:

- Silent Running (Naval)
- Reactor Engineering (Energy)

Enables:

- Atomic bomb delivery by submarine
-

Heavy Strategic Bombing Doctrine

Requires:

- Bombing Calibration (Air)
 - Mass Production (Industry)
-

Mechanized Warfare Doctrine

Requires:

- Breakthrough Doctrine (Land)
 - Strategic Rail Network (Industry)
-

Integrated Air Defense Network

Requires:

- Early Radar (Basic)
 - Radar Wave Theory (Energy)
 - Signals Interception (Intel)
-

Rocket Development Program

Requires:

- Advanced Ballistics (Energy)
- Mass Production (Industry)

PROGRAM LAYER (ENDGOALS)

These are national or late-game commitments.

Atomic Weapons Program

(Project Manhattan equivalent)

Requires:

Energy:

- Uranium Processing
- Heavy Water Chemistry
- Reactor Engineering

Facility:

- Nuclear Reactor (constructed & undamaged)

Delivery:

- Flying Fortress Capability
OR
- Advanced Submarine Propulsion

Multi-turn final development.

Germany – Wunderwaffen Program

Requires:

- Rocket Development
- Jet Engine Theory
- Mechanized Warfare Doctrine

Choice branch:

- V2 Campaign
 - Super Heavy Armor
 - Advanced Jet Interceptors
-

Germany – Atlantic Wall

Requires:

- Defensive Fortifications
 - Civil Defense Planning
 - Fleet Screening Doctrine
-

Japan – Kido Butai Doctrine

Requires:

- Fleet Screening Doctrine
 - Long Range Aircraft
 - Heavy Strategic Bombing Doctrine
-

United Kingdom – Global Intelligence Network

Requires:

- Cryptographic Systems
- Signals Interception
- Resistance Coordination

Synergy:

- Amplifies sabotage & resistance effects
-

Soviet Union – Deep Battle Doctrine

Requires:

- Breakthrough Doctrine
 - Industrial Relocation
 - Mechanized Logistics
-

What You Now Have

- Deep vertical progression
- Early incremental benefits

- Cross-branch strategic fusion
- Long nuclear path (5–7 rounds realistic)
- Intelligence as real strategic weapon
- National identity without commercial feel
- No single straight-line rush to the bomb

This is structurally complete.

If you ever want to refine this further, the next logical step would be:

1. Tier timing calibration
2. Exact mechanical effects
3. Research cost scaling
4. Anti-snowball balancing

But structurally?

This is solid.

And yes — Propaganda → Resistance was absolutely the right addition.

Descriptions:

Yes.

Now we turn structure into **game substance**.

Below are **clean, incremental, mechanically usable definitions** for every item in the system.

Design principles used:

- Every tech gives a *small, tangible effect*
- No tech wins the game alone
- Effects are modular and stack logically
- Nothing breaks G40 economy balance
- No wild power spikes

I'll keep wording rules-neutral so you can tweak numbers later.

LAYER 1 – BASIC INNOVATIONS

Improved Training

Infantry roll 1 die per round where one miss may be rerolled if attacking with artillery support.

Small reliability boost. No raw stat change.

Morale Systems

Requires: Improved Training

Once per turn, one land battle may ignore the first casualty taken.

Represents cohesion under fire.

Entrenchment Doctrine

Infantry defending in original territories may absorb 1 extra hit on a 1 before removal.

Makes homeland defense slightly stickier.

Standardized Parts

All newly produced units cost 1 IPC less once per turn (max 1 unit per turn).

Small industrial efficiency boost.

War Bonds Campaign

Roll 1 die at end of turn. On 3+, gain 1 IPC.

Stable minor economy boost.

Early Radar

AA guns hit on 1–2 instead of 1.

Small air deterrent.

Improved AA Procedures

Requires: Early Radar

Each AA gun may fire at +1 additional target.

Not stronger — just broader coverage.

Rail Coordination

You may move 1 additional land unit from a factory along connected territories.

Mobility bump.

Port Efficiency

New naval units may move 1 sea zone when placed (non-combat only).

Faster deployment tempo.

Basic Espionage

Once per turn, peek at one opponent's IPC total before purchases.

Information edge.

AIR DOMAIN

Improved Airframes

Aircraft defend at +1 on one die once per battle.

Small durability increase.

Engine Optimization

Requires: Improved Airframes

All aircraft gain +1 movement.

Mobility matters.

Long Range Aircraft

(Long range airplanes)

Aircraft may perform strategic bombing + combat in same turn if movement allows.

Operational flexibility.

Heavy Bomber Design

Heavy bombers roll 2 dice for bombing, choose best.

Controlled power.

Bombing Calibration

Strategic bombing ignores first 1 damage rolled.

Consistency.

Defensive Turrets

Heavy bombers may fire defensively when intercepted.

Survivability.

Strategic Payload Integration

Heavy bombers may carry special weapons (Atomic requirement).

Gateway tech.

Air Transport Doctrine

(Transport planes)

May airlift 1 infantry per turn (range limited).

Operational reach.

Jet Engine Theory

Requires: Advanced Metallurgy

Fighters attack at +1 first round only.

High-risk spike.

NAVAL DOMAIN

Improved Transport Capacity

(Improved Transports)

Transports defend at 1.

Heavy Transport Capacity

(Heavy Transports)

Transports carry 3 infantry OR 1 tank + 1 infantry.

Advanced Submarine Hulls

Subs defend at 2 instead of 1.

Extended Range Submarines

Subs gain +1 movement.

Silent Running

Subs may ignore first destroyer cancellation per turn.

Limited stealth.

Nuclear Submarine Doctrine

Requires: Reactor Engineering

Subs may move 2 extra in non-combat.

Strategic mobility.

Fleet Screening Doctrine

Destroyers give +1 defense to one capital ship.

Advanced ASW

Destroyers hit subs on 1–2.

LAND DOMAIN

Improved Artillery

(Advanced Artillery)

Artillery may support 2 infantry.

Artillery Fire Control

Artillery attack at 3 when paired.

Self-Propelled Artillery

Artillery move 2.

Mechanized Logistics

Tanks may blitz through 1 additional territory if unoccupied.

Breakthrough Doctrine

First round tank hits may not be assigned to tanks.

Encourages armored pushes.

Defensive Fortifications

Place 1 free bunker marker per turn in controlled territory (absorbs 1 hit in defense).

Urban Warfare Training

Infantry defend at +1 in cities/VC territories.

INDUSTRY DOMAIN

War Economy Planning

May convert 1 captured factory to operational next turn.

Industrial Mobilization

Factories produce +1 unit (max 1 factory per turn).

Mass Production

One unit type costs 1 IPC less this turn.

Advanced Assembly Lines

May build one extra aircraft per turn.

Supply Chain Optimization

Strategic bombing damage reduced by 1.

Strategic Rail Network

Move 3 land units from factory instead of 2.

Civil Defense Planning

Capital ignores first 2 SBR damage.

Industrial Relocation

Once per game move a minor factory.

ENERGY & PHYSICS

Advanced Metallurgy

Required for Jets, Rockets, Flying Fortress.

Uranium Processing

Allows reactor construction.

Heavy Water Chemistry

Reduces reactor build cost.

Reactor Theory

Enables Nuclear Reactor facility.

Reactor Engineering

Required for Atomic Program & Nuclear Subs.

Rocket Propulsion Theory

May build Rocket sites.

Advanced Ballistics

Rockets hit on 1–2.

Radar Wave Theory

All fighters intercept at +1.

INTELLIGENCE DOMAIN

Codebreaking

Once per game force opponent to reroll one battle.

Cryptographic Systems

Opponents must reveal one research choice.

Counterintelligence

Ignore one sabotage attempt.

Espionage Networks

Sabotage minor facility on 1–2 once per turn.

Scientific Espionage

On successful sabotage, steal 1 tech prerequisite.

Signals Interception

Reduce one enemy air movement by 1 once per turn.

Propaganda Campaign

Place influence marker in neutral territory.

Political Influence Operations

Reduce resistance cost by 1.

Resistance Coordination

Spawn 1 partisan in occupied original territory (once per turn).

Feeds UK Program.

CROSS-DOMAIN

Flying Fortress Capability

Heavy bombers ignore first interceptor hit.

Enables atomic delivery.

Advanced Submarine Propulsion

Subs may carry special weapons.

Heavy Strategic Bombing Doctrine

Add +2 to one bombing raid.

Mechanized Warfare Doctrine

Tanks attack at 4 first round only.

Integrated Air Defense Network

AA may reroll one miss per battle.

Rocket Development Program

Rockets may target airbases or naval bases.

PROGRAMS

Atomic Weapons Program

Multi-turn build.

Atomic attack destroys factory and all units in territory (one-use weapon).

Wunderwaffen Program

Choose:

- V2: Rockets hit capital automatically
 - Super Heavy Armor: 1 tank attacks at 5
 - Advanced Jets: Fighters defend at 5 first round
-

Atlantic Wall

All coastal territories gain bunker marker.

Kido Butai Doctrine

Carriers carry +1 aircraft.

Global Intelligence Network

Resistance actions cost 0 IPC.

Deep Battle Doctrine

Infantry may move after combat if territory cleared.

This is now mechanically complete.

Next stage — if you wish — would be:

- Remove 15–20% for balance
- Adjust IPC cost per research tier
- Make nuclear slightly harder
- Or fine-tune national asymmetry

But structurally and mechanically?

You now have a full Civilization-grade R&D system for G40.